

UNIT 1

Beware and take care

The junzi always thinks about potential disasters in order to guard against them beforehand.

—The Book of Changes

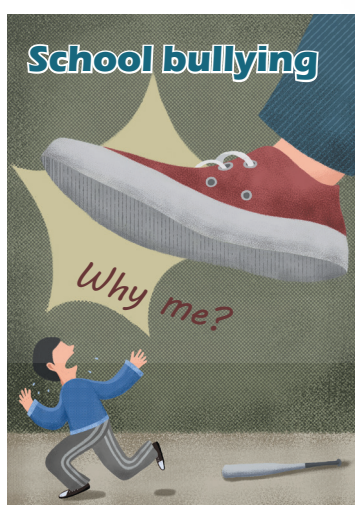


In this unit, you are going to:

- read a feature article about a highly infectious disease;
- write an article calling on people to look out for cybercrime;
- read a speech transcript on drugs;
- make a booklet on safety education.

Welcome to the unit

As social beings, we are faced with different kinds of social issues. Look at the posters below and discuss the following questions in pairs.



- 1 Which of the three posters above is the most striking to you? Why?
- 2 What other slogans can you think of for the posters?
- 3 What can you do to help prevent these problems?

Reading



Infectious diseases, such as malaria and flu, have long been part of human history. The feature article below describes the fight against a certain type of infectious disease. Before you read the article, think about the following questions:

- What do you know about infectious diseases?
- What can you personally do to fight infectious diseases?

Fighting the hidden enemy

An enemy appeared without warning, attacking its unsuspecting victims with efficiency. This silent and invisible enemy swiftly moved through communities and secretly crossed borders. The world soon learnt that the enemy we faced was Covid-19.

5 Covid-19 is caused by a new strain of coronavirus. Once entering the body, the virus multiplies inside the cell quickly and infects neighbouring cells. Common symptoms can include fever, tiredness, a dry cough, a headache and loss of taste or smell. However, in some cases, the virus causes the body's own immune system to overreact, resulting in the body killing its own healthy cells. Consequently, the disease can become worse, causing damage to the body's organs and tissues.

10 Covid-19 is a highly infectious disease that can be transmitted easily and swiftly from person to person. This typically happens when people take in respiratory droplets and particles from an infected person. They can also catch Covid-19 by touching a surface or an object contaminated with the virus, and then touching their own mouth, nose or eyes. Research shows that there are certain settings that allow the virus to spread more
15 easily. For instance, it favours cool, dry conditions and crowded indoor spaces with poor ventilation. What makes this disease even more dangerous is that infected people who show no symptoms may become unknowing carriers and spreaders of the virus.

In early 2020, the threat posed by Covid-19 came to be fully recognized on a global scale. It spread rapidly across the world, affecting millions of people. More dangerously,

20 new and even more infectious variants of the virus, such as the Delta and Omicron variants, gradually began to emerge in various parts of the world, presenting an increased threat to the health of the world's population.

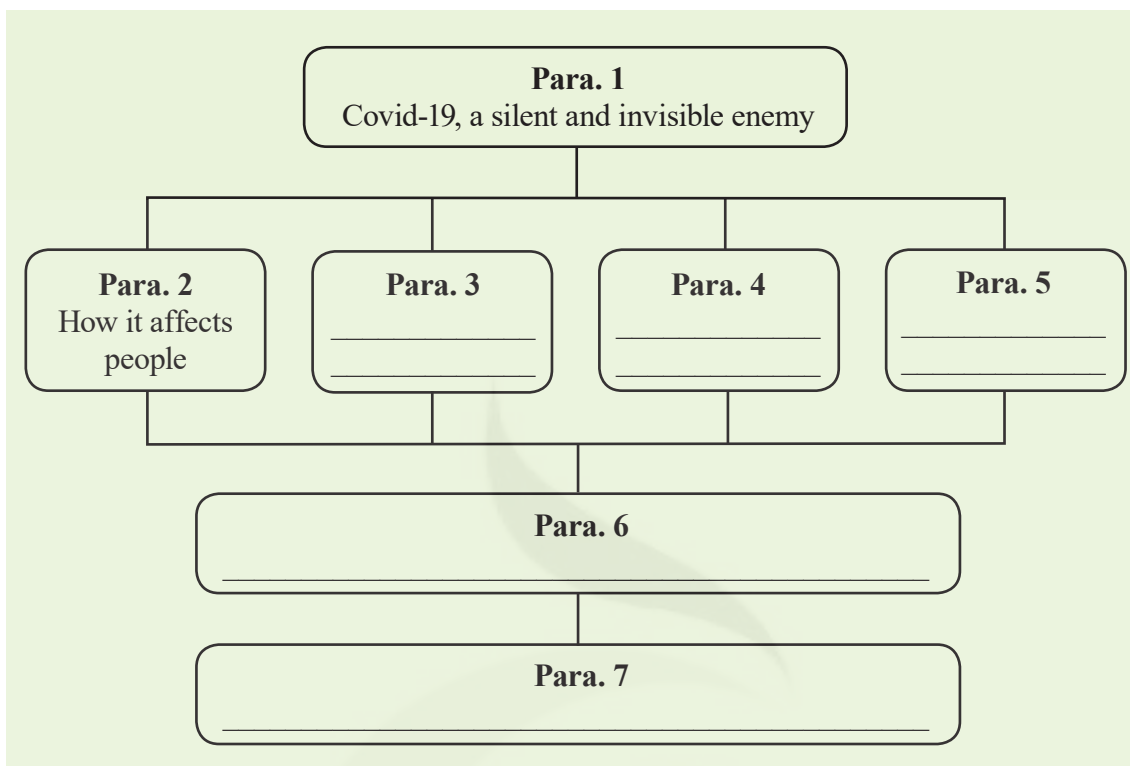
Fortunately, the world pulled together to beat this silent and invisible enemy. Governments developed and pursued clear policies in response to the sudden outbreak
25 of the pandemic. For example, the Chinese government constantly followed the principle of “putting the people and their lives above all else”. Governments also joined forces with leading research institutes so that vaccines could be developed, produced, distributed and administered in record time. Doctors and nurses worked round the clock and went wherever they were needed, risking their own lives to tirelessly treat and care
30 for patients. Community workers and volunteers stepped forward to deliver food and medicines to residents. Ordinary people from all walks of life united in helping those affected by the pandemic. Simple acts of kindness and generosity were everywhere. In the darkest days when Covid-19 was doing its worst, the human spirit shone at its brightest.

35 Without doubt, Covid-19 is just one of the many infectious diseases that we have encountered in human history. While humankind has made great progress in the fight against various types of infectious diseases, we still cannot live in a world completely free of them. However, we can live more safely if we act wisely. There are preventive measures that everyone can take to help control the spread of infectious diseases.
40 Firstly, maintaining personal hygiene is essential. Clean your hands with soap and water thoroughly. Avoid touching your mouth, nose and eyes with unwashed hands. Secondly, when in public places, you can wear a face mask to protect yourself from viruses. Thirdly, you can receive vaccines to help your body build up immunity against certain infectious diseases. For example, receiving an annual flu vaccine can lower the risk of
45 getting the flu. Finally, seek medical attention if you believe that you have been exposed to some type of virus. By staying alert and following professional advice, we not only protect ourselves, but also help protect our fellow citizens.

The long history of our fight against infectious diseases has shown us that we should live in a human community with a shared future. Only by working together, while accepting
50 our individual responsibilities, can we solve both national and global problems. We need to support each other so that we can march confidently, arm in arm, into a better and more secure future.

A Understanding the text

A1 Read the feature article and complete the chart below with the main idea of each paragraph.



A2 Read the feature article again carefully and answer the following questions.

1 What happens once the virus enters the body?

2 How is Covid-19 spread from person to person?

3 What did community workers and volunteers do to fight Covid-19?

4 What should we do in public places to help control the spread of an infectious disease?

A3 In pairs, discuss the following questions.

1 Why does the author compare Covid-19 to “the hidden enemy” in the title?

2 The author writes, “In the darkest days when Covid-19 was doing its worst, the human spirit shone at its brightest.” What examples can you think of that support this point of view?

B Building your language

B1 The website article below is about the flu. Complete the article with the correct forms of the words and phrases in the box below.

symptom
pose

gradually
lower

infectious
result in

build up
in record time

In 1918, a silent and invisible enemy struck the world. Nobody knows exactly where it originated, but ⁽¹⁾ _____ it found its way into nearly every corner of the globe, like a hurricane across the open fields. It swept across the globe in three waves, and claimed the lives of an estimated 50 million people. The cause of this huge loss of life was a strain of the H1N1 virus, ⁽²⁾ _____ the “Great” flu pandemic of 1918–1919.

Generally, the flu is a(n) ⁽³⁾ _____ disease of the respiratory system. At first, the flu can appear like the common cold with a dry cough, a runny nose and a sore throat. Very quickly, ⁽⁴⁾ _____ can include fever, aching muscles and a headache. Nowadays, most people ⁽⁵⁾ _____ recover within a week. However, the flu can ⁽⁶⁾ _____ a serious threat to high-risk groups such as the elderly and those with underlying health conditions. Flu viruses can be spread when people breathe in the droplets from an infected person or touch everyday surfaces or objects which have the flu virus on them.

To stop the virus from spreading, you should take some preventive measures. Use tissues to cover your mouth and nose when coughing or sneezing, and clean surfaces or objects regularly. Also, wash your hands with soap and water frequently. If you had the flu in the past, your body would ⁽⁷⁾ _____ immunity to that specific strain of the flu virus. Unfortunately, immunity reduces over time and flu viruses continue to evolve, so it would be wise to receive an annual flu vaccine. While the flu vaccine is not one hundred per cent effective, it can ⁽⁸⁾ _____ the risk of having severe problems.

Since the flu comes and goes, we need to always pay close attention to the prevention of it. Remember that “prevention is better than cure”!

B2 The feature article uses a scene to introduce the topic of Covid-19, which helps grab the reader’s attention. Find the example in the article and write the opening paragraph of an article about the impact of seasonal flu on your life.

Grammar and composition

Linking words

A Exploring the rules

Below is a blog post written by Kevin about a self-medicating experience that put him at risk. Find the different categories of linking words and fill in the table below. The first one has been done for you.

It is said that we should look after our body as it is the only place we have to live in. However, I wish I had paid closer attention to this and taken good care of my body.

I woke up one morning with a bad backache and I assumed that it was because I had pulled a muscle in my back when playing basketball the day before. Rather than do the sensible thing of going to see a doctor, I searched the Internet for suitable painkillers and ordered some online. For one thing, I was too lazy to drag myself to the hospital. For another, I thought I knew what I was doing. What a mistake! I ended up making myself seriously ill. After taking the painkillers for a few days, I began to lose my appetite and feel tired. Not only did my back pain get worse, but I also started having a stomach ache. Then it turned out that my back pain had nothing to do with a muscle strain. On the contrary, it was caused by a serious liver infection. As I learnt later from my doctor, I had taken too many painkillers, which were irritating the lining of my stomach. Additionally, these particular medications were damaging an already damaged liver! As a result, I had to spend two weeks in hospital.

In the end, I made a full recovery, but it was a costly and painful learning experience. It is often said that a little bit of knowledge can be a dangerous thing, and this was certainly true in my case. I learnt the hard way that when it comes to our own health, we should seek professional medical advice at the first sign of trouble. Remember, our body is our most priceless possession.

Sequence	
Contrast	however
Cause and effect	
Addition	

Working out the rules

- We use sequence words like *first*, *then* and ⁽¹⁾ _____ to show the order of events.
- We use contrasting words like ⁽²⁾ _____, *on the contrary* and *on the other hand* to introduce an idea in contrast to the previously mentioned one.
- We use cause-and-effect words like *for one thing ... for another*, *therefore* and ⁽³⁾ _____ to introduce reasons or results.
- We use addition words like *besides*, *moreover* and ⁽⁴⁾ _____ to introduce additional information.

🔗 Grammar and composition notes → page 94

B Applying the rules

B1 Rewrite the following sentences with the linking words in the brackets.

- 1 When there is flu going around, you should learn to protect yourself. You need to wear a face mask. (first of all)

- 2 During the fire drill, students are required to bend down and leave through the emergency exit. They need to cover their mouth and nose with a wet towel. (furthermore)

- 3 At an intersection without zebra crossings, do not cross the road directly. You can use a flyover or an underpass to cross the road. (instead)

B2 Below is an online article written by a medical expert giving health care tips. Complete the article with the correct linking words in the box below.

moreover although last but not least however then as a result

Sometimes we self-medicate. Maybe we believe the problem is not that serious and that we will feel better tomorrow, or perhaps we just cannot find the time to go to hospital. (1) _____, we often do a quick search online and reach for medicines such as cough mixture and headache tablets without a second thought.

However, self-medication could have disastrous consequences. As symptoms of different illnesses can be very similar in the initial stages, a non-professional may misdiagnose an illness. (2) _____, you are likely to take an inaccurate dose of medicine. Self-medication may also lead to delays in seeking medical advice when it is most needed. To avoid the trap of self-medicating, it is absolutely essential to go to hospital and have yourself examined by a doctor with professional skills and specialist equipment.

Once a professional diagnosis has been made, you may be prescribed a course of treatment. (3) _____ you need to take the medication exactly as advised by your doctor. Ensure that you know the answers to the following questions: what is the recommended dose? How many times a day should you take it? How long should the treatment last? What are the possible side effects? Are there any dangerous interactions with other drugs or with certain foods? (4) _____ the amount of information that you need to keep track of might sound excessive, not following a course of treatment properly may result in the illness coming back even more strongly, or lead to other health conditions.

(5) _____, I am sure we all know that prevention is better than cure. Therefore, to minimize the need for a visit to the doctor, ensure you keep yourself healthy by eating well, exercising regularly and maintaining high standards of personal hygiene. (6) _____, once you feel ill, you should avoid self-medicating and get immediate medical help.

B3 Supposing someone you know is misusing medicine, write a short letter of advice to him or her. Use proper linking words in your writing.

Integrated skills

Fighting against cybercrime

A Joey is talking to his friend Emily about a recent experience. Listen and finish the exercises below.



A1 Listen to the conversation and choose the best answer. Circle the letters of the correct options.

- 1 What was the original price of the guitar?
a 400 yuan. b 800 yuan. c 1,600 yuan.
- 2 How much money did Joey lose in total?
a 800 yuan. b 2,500 yuan. c 5,000 yuan.
- 3 Why is QR code fraud on the increase?
a Because more people are not very careful with online payment.
b Because QR codes are widely used.
c Because fake QR codes are seen in all places.
- 4 Where do criminals often put fake QR codes?
a On websites and in smartphone apps.
b In shops and restaurants.
c At airports and train stations.



A2 Listen to the conversation again and complete the notes below.

QR code fraud

Joey's experience

- He saw ⁽¹⁾ _____ for a guitar at half price.
- He scanned a QR code to pay.
- His bank details were stolen, his ⁽²⁾ _____ was cleaned out and he lost all the money.
- He reported the fraud to the police as soon as he found out.

Emily's ⁽³⁾ _____

- Stay alert. Remember that if something seems ⁽⁴⁾ _____, it probably is.
- Check a QR code carefully.
- When using a QR code to make a payment, pay close attention to ⁽⁵⁾ _____ before you confirm the payment.
- Install and register the National Anti-fraud Center app.

B Joey has found a website article about cybercrime. Read the article below and answer the following questions.

Cybercrime: the dark side of the Internet

It was estimated that as of January 2021, there were 4.66 billion active Internet users globally, representing almost 60% of the world's population. The Internet has, in a relatively short span of time, completely revolutionized how we communicate with each other, exchange information and conduct business. Unfortunately, due to its wide reach and relative privacy, the Internet has become very attractive to criminals. Consequently, cybercrime has become a huge problem in recent years. As most of us now use the Internet as part of our daily life, an awareness of cybercrime is essential, if we want to avoid becoming a victim.

Cybercrime can come in many different forms. Phishing is one of the most common types of cybercrime. When clicking on links to dangerous websites or software, you may be fooled into providing personal information such as passwords, usernames or financial details. Malware, another form of cybercrime, can cause damage or gain illegal access to a computer or network. Cybercriminals trick you into parting with your money, infecting your computer with a virus to steal your personal data, or blocking you from accessing your own data until a ransom is paid. For example, in 2021, a computer manufacturer suffered a ransomware attack. Cybercriminals took advantage of a weakness in a server, accessed the company's files and demanded a large sum of money.

Fortunately, you can take preventive measures to reduce your chances of falling victim to such cybercrimes. Firstly, you need to stay alert when using the Internet. Do not provide personal information on a website you know nothing about, especially those that ask for your name, mailing address and bank account number. To avoid getting taken in, be cautious about advertisements offering bargains and rewards that seem too good to be true. And, of course, do not open email attachments or click on links from unknown sources, no matter how curious you might be! Secondly, you can turn to technical defences to protect yourself from cybercrime. Turn on your spam blocker to prevent unwanted messages from getting to your inbox. Make sure you have reliable antivirus software and always keep it turned on. It is also a good idea to keep your software and operating system up to date by downloading and installing the latest security patches.

So, next time you go online, never drop your guard. Stay alert, smart and safe!

1 How do you understand that cybercrime is “the dark side of the Internet”?

2 What preventive measures should we take to avoid falling victim to cybercrime?

C In pairs, discuss the problem of cybercrime. Use the following questions and expressions to help you.

- What is cybercrime?
- What types of cybercrime are there? Explain one type with a specific example.
- What impact does cybercrime have on victims?
- How can we protect ourselves from cybercrime?

Expressions

Expressing concerns

There is growing concern that has caused widespread concern.

The public are increasingly concerned about is of great concern to ...

It greatly concerns/worries people that ...

D Write an article calling on people to look out for cybercrime. Use your ideas from part C and the information in parts A and B to help you. Check and improve your writing after you finish it.

Title: _____

Introduction
to cybercrime

Types of
cybercrime
and the
impact

Ways to
prevent
cybercrime

Conclusion

Extended reading



Read the transcript of a speech given by a police officer to senior high students about drugs.



Good morning, everyone. As a police officer who has spent many years working in the front line of the fight against drugs, I have witnessed, all too often, the destructive impact that drugs can have on individuals, families and the wider community. I have seen how drugs can destroy people's physical and mental well-being and ruin young
5 lives before they even begin. I have seen how families are torn apart after a family member has become addicted to drugs. I have also seen how tirelessly the forces of law and order work together in fighting against drugs. I will not exaggerate the scale of the problem, as the facts speak for themselves. According to the 2021 World Drug Report, it is estimated that around 275 million people used drugs worldwide and over 36
10 million people suffered from drug use disorders in the last year. These shocking figures indicate that drug abuse is a very serious global problem. Therefore, it is everyone's responsibility to say no to drugs.

Drug abuse hurts an individual drug user both physically and mentally. The most serious health conditions associated with drug abuse include heart problems, respiratory
15 problems, kidney failure, liver damage and a weakened immune system, increasing the risk of illness and infection. Moreover, drugs interfere with a person's ability to make sensible decisions. Consequently, people under the influence of drugs are no longer in control of their own mind and likely to put themselves or others in great danger. They also suffer from memory loss and mood disorders like depression or anxiety.

20 Apart from its harm to individuals, drug abuse also destroys families and impacts society as a whole. Drug users sometimes lie and steal to fuel their habit, while ignoring the responsibilities they have to their families. This can lead to not only financial hardship but also the breakdown of once strong family relationships and the loss of

stability within the family. The impact drug abuse has on society is also enormous.

- 25 Under the influence of drugs, drug users may exhibit violent behaviour and sometimes even commit criminal acts, threatening public safety. They are willing to do anything to get their hands on drugs. Also, because drug users need a regular supply of drugs, the criminal underworld quickly develops, eating away at the fabric of society.

- 30 Knowing the serious damage that can be caused through the use of drugs, you may be asking yourself: why would anybody be foolish enough to start taking drugs? There is no simple answer to this question, but there are well-identified risk factors for drug abuse. Some people experiment with drugs simply out of curiosity, some regard drug use as an escape from the stresses of everyday life, and others start to use drugs just because people around them are drug users.

- 35 Fortunately, there are things you can do to protect yourself from drug use. First of all, drug education is of vital importance. You should learn more about drugs, especially new designer drugs, which are incredibly dangerous. The more you know about the terrible consequences of taking drugs, the more likely you are to stay away from them. Secondly, you need to choose where you go and who you spend time with very carefully.
- 40 Make sure you surround yourself with friends who you can really count on and turn to for support. Last but not least, should you ever come under pressure to use drugs, you must immediately seek assistance from an adult who you look up to, such as a family member, a teacher or a doctor.

- 45 In conclusion, I hope my message is clear—say no to any kind of drug. The most effective way to avoid all the problems caused by the use of drugs is not to use them in the first place.

A Write a summary of the speech transcript. Use the following ideas to help you.

- Dangers of drug abuse
- Reasons for using drugs
- Ways to protect yourself from drug use

B How can teenagers receive an effective drug education? Support your opinion with details.

Project

Making a booklet on safety education

A As a class, discuss different safety issues you are interested in. You can use the ideas below or think of other ideas. Then in groups, choose one safety issue to research.

health safety
fire safety

food safety
information safety

road safety
electrical safety

B As a group, research your chosen safety issue. Use the questions below to help you.

- What is the chosen safety issue?
- What are the dangers?
- How can we guard against them?

C As a group, put together your information to make your part of the booklet. Use the example below to help you. Present your part to the rest of the class. Then as a class, put all the parts together to make the booklet.



HIV doesn't care, but you must

◆ What is HIV?

HIV is a virus that attacks the body's immune system. If left untreated, HIV can cause AIDS, the late stage of HIV infection, meaning that the body's immune system has become so weak that it cannot fight off certain kinds of infections. Unfortunately, there is no effective cure for HIV yet.

◆ What are the early symptoms?

Someone who has HIV may develop flu-like symptoms within two to four weeks after infection. These symptoms may last for a few days or several

weeks. Possible symptoms include:

- Fever
- Chills
- Rashes
- Muscle aches

However, many people may not show any symptoms in the early stages of HIV infection. The only sure way to know is to get tested for HIV.

◆ How can you stay safe?

To stay safe, you should:

- avoid unprotected sex;
- not share needles;
- seek medical advice immediately if you think you have been exposed to HIV.

Assessment

Complete the following learning journal to assess your performance. Then work in groups and exchange your learning journal with your partners.

Learning journal

In this unit I learnt _____

I was most interested in _____

I did well in _____

However, there is something I need to improve on: _____

In order to make more progress, I will _____

* Assess your learning of vocabulary and grammar by doing language practice on pages 57–58.

Further study



The world is faced with a wide range of social problems. Some problems are world issues such as climate change and pollution, while others are of a more individual nature, for example health and family matters. Choose a social problem that concerns you, big or small, and search online to learn more about the topic.



The outbreak of Covid-19 has changed various aspects of our lives on a global scale, including health care, education and lifestyle. Listen to a talk about Covid-19 to learn more about it.

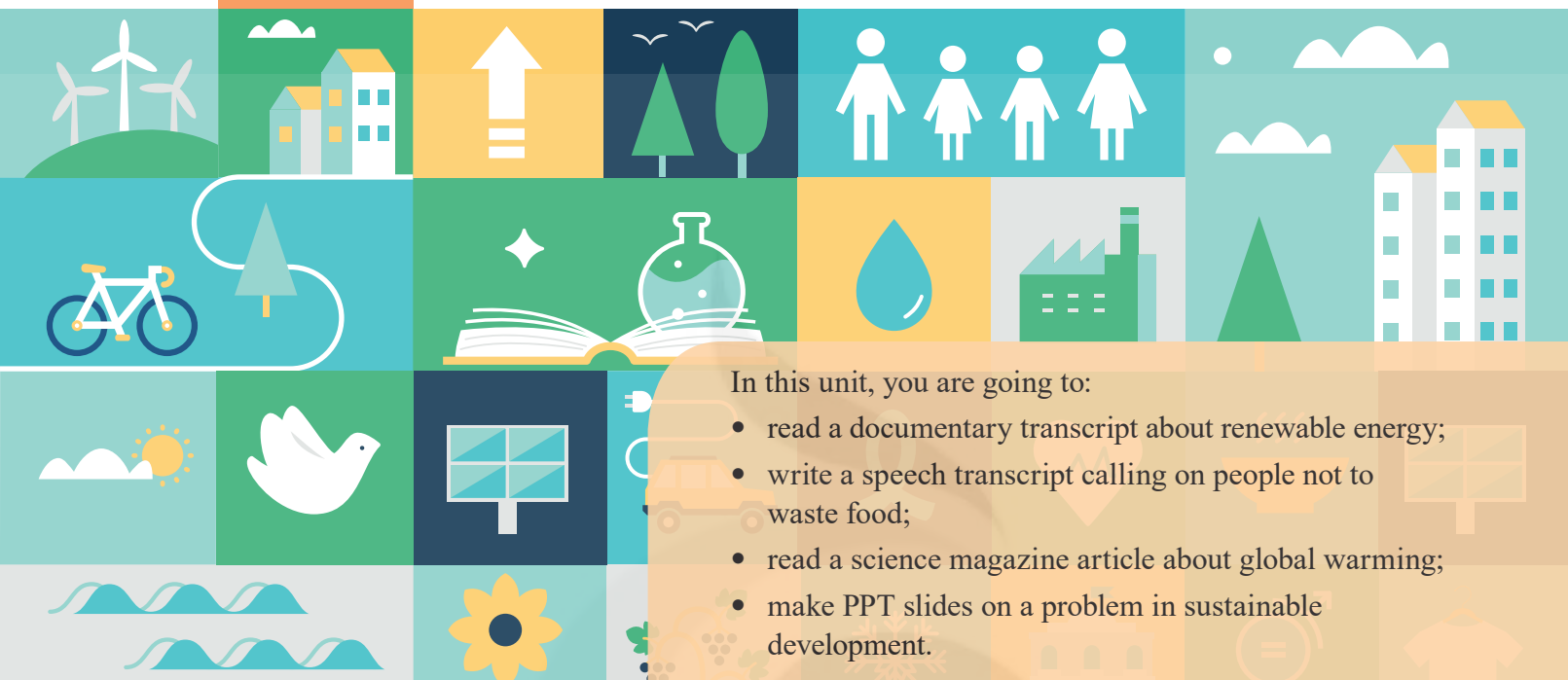


UNIT 2

Building the future

Sustainability is about ecology, economy and equity.

—Ralph Bicknese



In this unit, you are going to:

- read a documentary transcript about renewable energy;
- write a speech transcript calling on people not to waste food;
- read a science magazine article about global warming;
- make PPT slides on a problem in sustainable development.

Welcome to the unit



Sustainable development is a major global trend that will transform our world. Watch the video and finish the following exercises.



Three core elements:

- _____
- _____
- Environmental protection

Three SDGs:

- _____
- _____
- Sustainable cities and communities

- 1 How do the three Sustainable Development Goals mentioned in the video target the three core elements?
- 2 What other goals do you think we need to achieve to ensure sustainable development?

Reading



Traditional energy sources, such as coal, gas and oil, have caused problems for our planet, so we need to find new types of sustainable energy. The documentary transcript below describes renewable energy sources. Before you read the transcript, think about the following questions:

- What do you know about the environmental impact of coal, gas and oil?
- What renewable energy sources have you heard about?



There was once a village surrounded by fertile fields with farmers tending their crops and farm animals grazing lazily below a mighty mountain. Then coal was discovered, and soon titanic-sized earth-digging machines circled the village. Villagers eagerly accepted the high-paid jobs the mining company provided, bringing a temporary economic boom. Meanwhile, the machines cut through everything in their path to access the coal buried beneath. Many years later, all that remained was a deserted village and an enormous hole.

The scenario is tragic, but it only partly describes the damaging effects fossil fuels have on the environment when they are mined. Actually, once burnt, they produce a large quantity of carbon dioxide, which in turn traps heat and causes global warming. Humans are clearly facing a critical moment, so a move towards clean, renewable energy is inevitable. Renewable energy sources, which include solar, wind, hydroelectric, tidal, biomass and geothermal energy, are perfect alternatives to fossil fuels. One of the reasons is that their carbon dioxide emissions tend to be lower. A study found that renewable energy sources emit about 50g or less of carbon dioxide per kilowatt-hour over their lifetime, compared to about 1,000g for coal. Besides, fossil fuels are non-renewable, whose reserves are estimated to dry up one day in the future. In contrast, renewable energy sources are basically inexhaustible and will be available to us for a few billion more years if our planet remains safe and healthy.

Countries around the world are racing towards a low-carbon future by investing in renewable energy sources. Renewable energy is now gradually being integrated into infrastructures, such as transport, communication and power supply systems. China is one of the countries embracing the renewable energy revolution. China manufactures

state-of-the-art solar panels and wind turbines, and is the global leader in the production
25 of batteries to power electric vehicles and store renewable energy on power grids.
Renewable energy consumption accounts for more than 20 per cent of its total energy
consumption, and this figure is increasing. Over the past decade, China has been ahead in
the development and application of clean energy technologies.

So, with these advantages and developments, why not completely replace fossil fuels
30 with renewable energy? The truth is that producing power from renewable energy
sources can be challenging. Renewable energy sources are not available everywhere and
at any given moment. For instance, tidal energy is only available in coastal areas; solar
and wind energy cannot produce power 24/7. A greater challenge is storing any extra
energy in batteries for later use. Most current batteries do not have a storage capacity
35 large enough to meet our daily power demands. Certain high-capacity batteries present
problems, such as high manufacturing costs and related environmental damage. For
instance, mining for the rare metals used in the batteries is expensive, and when not
properly recycled, the batteries leak dangerous chemicals into the environment.

With these challenges in mind, a future with renewable energy may be difficult to
40 predict. However, it is possible. Renewable energy sources exist in all kinds of places,
including space. Space-based solar power projects aim to collect solar energy in outer
space and distribute it everywhere it is needed on the Earth. A solar collector would be
placed in a location where there is constant sunlight to capture higher levels of solar
energy for longer periods. Scientists are also working hard to solve the energy storage
45 problem. Thermal batteries, for instance, can store the extra renewable energy as heat
up to the extremely high temperature of 1,000°C in an inexpensive storage medium
like sand or water. Compared with common batteries, these thermal batteries are
considerably cheaper, and have a longer life as well as a higher storage capacity. Better
still, they are completely recyclable.

50 Without doubt, renewable energy will form a big part of our future. Looking forward, we
should invest in new technologies and innovative thinking that open the door to a whole
new world where we decrease reliance on fossil fuels. Renewable energy—a welcome
gift for our future—will help us build a greener and healthier world.



A Understanding the text

A1 Read the documentary transcript and complete the chart below with the main idea of each paragraph.

Introduction	Para. 1: Fossil fuels have damaging effects on the environment.
Main body	Para. 2: _____
	Para. 3: _____
	Para. 4: _____
	Para. 5: _____
Conclusion	Para. 6: Renewable energy is a gift for our future.

A2 Read the documentary transcript again carefully and answer the following questions.

1 What types of renewable energy sources are mentioned in the transcript?

2 In what aspects has China been ahead over the past decade?

3 What problems do certain high-capacity batteries present?

4 How can thermal batteries solve the storage problem of renewable energy sources?

A3 In pairs, discuss the following questions.

1 The author says renewable energy is a gift for the future in the title. Why does the author say that? Use details from the transcript to support your idea.

2 The author uses a tragic scenario to introduce the topic of renewable energy. What do you think of this introduction?

3 In paragraph 3, the author says, “China is one of the countries embracing the renewable energy revolution.” What examples in China do you know of that support this statement?

B Building your language

B1 The magazine article below is about zero-energy buildings. Complete the article with the correct forms of the words and phrases in the box below.

solar
integrate

extra
estimate

trap
decrease

open the door to
in turn

Rapid urbanization has increased building construction and hence energy consumption. Traditional architectural designs contribute to depleting the Earth's natural resources, which (1) _____ accelerates global warming. The revolutionary concept of zero-energy buildings helps tackle the problem. To be considered as zero-energy, a building consumes energy that should equal its energy generated on site from renewable energy sources over a specific period. The structure may (2) _____ a variety of renewable energy systems with smart architectural designs, making it energy self-sufficient and energy-efficient.

The Centre for Sustainable Energy Technologies in Ningbo, Zhejiang Province, is an example of a zero-energy building. Among its various green technologies, the many (3) _____ panels on and around the building play a major role in producing power. On sunny days, the panels can generate 600 to 800 kilowatt-hours of power. Any (4) _____ power is stored, providing the building with electricity once the sun sets. Moreover, the building's smart designs include a double skin of glass on the south. The design optimizes sunlight and minimizes the need for artificial lighting. Besides, it can (5) _____ warm air between the layers in winter and protect the building against excessive exposure to sunlight in summer. This design contributes greatly to (6) _____ power costs for lighting, heating and cooling. Notably, with all its green technologies and smart designs working together, the building (7) _____ to save more than 400 tonnes of coal and reduce carbon dioxide emissions by about 1,000 tonnes over 25 years.

Zero-energy buildings are a trend well worth following. China is implementing plans to reduce the number of energy-inefficient buildings across the country. With their low environmental impact, zero-energy buildings (8) _____ sustainable cities for the future.

B2 The documentary transcript uses point-by-point contrast when discussing the benefits of renewable energy. Find examples of such information in the transcript. Then write a short paragraph to compare one renewable energy source with a traditional one using point-by-point contrast.

Learn this Point-by-point contrast is the comparison of a specific point of an idea with that of another related idea. This method focuses on the differences between the two ideas. Using this method increases your understanding of the two ideas which are being compared and helps you identify connections between them.

Grammar and composition

The elements of a paragraph

A Exploring the rules

Below is an article about white pollution. Identify the elements of paragraphs 2 and 3 and fill in the table below. The first one has been done for you.

Annually, Mount Qomolangma draws many mountaineers, some of whom throw away rubbish, including plastic waste, when they leave. The Mariana Trench is not immune to plastic either. A plastic bag was found lying at its bottom. All this highlights an enormous problem: white pollution.

(1) Inevitably, white pollution poses serious threats to the environment. (2) Since its mass production in the early 1950s, only 9 per cent of plastic from the over 9.2 billion tonnes produced has been recycled, leaving the majority of plastic lying around. (3) With the decomposition process that can take decades or centuries, plastic slowly releases poisonous chemicals into the soil and groundwater. (4) Essential soil properties are negatively affected, and water quality is degraded by contamination. (5) If the problem of plastic pollution is not addressed, it will certainly damage our environment permanently.

(6) White pollution also exposes all living things to danger. (7) Plant growth is potentially affected, and animals experience various illnesses. (8) Moreover, animals suffer other consequences of consuming plastic items or becoming trapped. (9) It is estimated that more than 1 million seabirds and 100,000 marine animals die from plastic pollution every year. (10) As plastic travels through the food chain, its impact reaches the very top: humans are consuming food contaminated by white pollution. (11) Unbelievably, 1 in 3 fish caught for human consumption contains plastic. (12) Ingesting poisonous chemicals released by plastic is directly linked to certain diseases like cancers and immune system problems. (13) It appears that no living thing is safe from plastic pollution.

Fortunately, the problem can be tackled through reducing, reusing and recycling plastic products. Strictly implementing the above measures would require cooperation and participation from all sectors of society.

	Para. 2	Para. 3
Topic sentence	(1)	
Supporting sentences		
Concluding sentence		

Working out the rules

- A topic sentence clearly (1) _____ (states/supports) the main idea of a paragraph.
- Supporting sentences help (2) _____ (explain/restate) the main idea with details and specific examples.
- A concluding sentence (3) _____ (repeats/introduces) the main idea of the paragraph, but in a different way.

👉 Grammar and composition notes → pages 94–95

B Applying the rules

B1 Choose the proper topic sentence for each paragraph below. Write the correct letters in the blanks.

- 1 _____ We should avoid using straws where possible by choosing non-straw drinking methods, for example, using cups with straw-free lids. If we have to use straws, we could select those made of eco-friendly materials such as paper, bamboo or metal. Our choices have more impact than we think.
- a Reducing the use of plastic straws requires us to change our habits.
 - b Plastic straws are causing harm to the environment.
- 2 _____ This type of pollution disrupts people’s sleep patterns, leading to various health problems. It also affects animal behaviour. Furthermore, much of the artificial light at night is unnecessary and thus creates a massive waste of energy. The problem of light pollution must be addressed.
- a Light pollution is a serious environmental issue.
 - b Light pollution is a problem in major cities.

B2 Below is an article about a solution to white pollution. For paragraphs 2–4, put the sentences in the correct order and identify the elements of each paragraph.

Every year, people all around the world throw away billions of plastic spoons, forks and knives. These single-use plastic items wind up in landfills. Motivated to create something that does not pose risks to the environment and living things, an Indian company is offering a creative solution to this problem: eatable spoons!

(1) And if you don’t want to eat it, you can throw it into the compost. (2) These eatable spoons, invented by an Indian researcher, offer a waste-free alternative to traditional plastic spoons and are just as functional. (3) When you are done with the spoon, you do not throw it out—you eat it instead! (4) This significantly cuts down on the amount of plastic people use. (5) The spoon will decompose in just a few days once used.

(6) Next, they shape the dough into spoons and bake them. (7) Making the spoons is fairly simple. (8) First, a dough is made out of a blend of flour. (9) Finally, when the spoons come out of the oven, they are ready to use. (10) The eatable spoon has a shelf life of up to two years and is firm enough for hot liquids like soup and frozen foods like ice cream.

(11) The future of these spoons looks bright and delicious! (12) This way, you can use sweet spoons when you are eating dessert and salty ones when you are eating your main dish. (13) It plans to offer sweet and salty spoon options. (14) The company is still exploring the potential for these spoons.

	Para. 2	Para. 3	Para. 4
Order			
Topic sentence			
Supporting sentences			
Concluding sentence			

B3 In pairs, discuss more ways to reduce pollution. You can focus on one particular type, like air pollution or light pollution. Write a short paragraph that includes a clear topic sentence, some supporting sentences and a concluding sentence.

Integrated skills

Fighting food waste

A Gregory is listening to a podcast about food waste. Listen and finish the exercises below.



A1 Listen to the podcast and choose the best answer. Circle the letters of the correct options.

- 1 Why does the speaker use restaurants in America as an example?
 - a To tell us something about American restaurants.
 - b To let us know how food is wasted in American restaurants.
 - c To show the seriousness of food waste in American restaurants.
- 2 What requires large amounts of energy?
 - a Dealing with solid waste in landfills.
 - b Producing and transporting food.
 - c Breaking down uneaten food.
- 3 How powerful is the greenhouse gas produced when food breaks down in landfills?
 - a Up to 9 times more powerful than carbon dioxide.
 - b Up to 17 times more powerful than carbon dioxide.
 - c Up to 86 times more powerful than carbon dioxide.



A2 Listen to the podcast again and complete the notes below.



Food waste

Situation

- (1) _____ people suffered from hunger.
- More than (2) _____ tonnes of food were wasted.

Causes

- Imperfect fruit and vegetables are often rejected at retail stores.
- Restaurants (3) _____ and customers overorder.
- Private households waste food mainly because of (4) _____.

Negative effects

- It impacts the environment negatively: wasting energy and contributing to (5) _____.
- It makes (6) _____ more severe, leading to health problems such as malnutrition.

B After listening to the podcast, Gregory has found an article about the “Clean Plate” campaign to fight food waste. Read the article below and answer the following questions.

“Clean Plate” campaign: getting serious about food waste



In the burning sun, they hoe the crops.
Down into the earth sweat drops.
Who knows every grain on the plate
Is gained by taking pains great!
("Sympathy for the Farmers II")

In ancient times, this well-known Chinese poem encouraged people to save food, which was already a traditional Chinese virtue at that time. Now, the “Clean Plate” campaign is doing the same thing, especially as food waste has become a growing concern.

The “Clean Plate” campaign, which urges people to order, cook and eat the exact amount of food they need, kicked off in Beijing in 2013. It was quickly popularized across the country and was embraced wholeheartedly by the nation. The campaign is intended to raise public awareness of valuing food and educate people to save food. Eye-catching slogans and easy-to-understand information inspire citizens to support it enthusiastically. Many communities encourage their residents to adopt the concept of food conservation in their daily lives, ensuring it becomes a conscious practice. Meanwhile, many schools host activities to highlight the “Clean Plate” campaign and issue official proposals that encourage students to support the campaign both at school and at home.

In April 2021, the *Law of the People's Republic of China on Food Waste* took effect, which aims to prevent food waste, safeguard food security and promote the traditional virtues of the Chinese nation. Apart from making a law at national level, successfully applying the “Clean Plate” measures also requires everyone to do their part. Some canteens and restaurants, aided by big data technology, can estimate the amount of food needed for large groups, preventing food waste. Moreover, they are preparing smaller serving sizes, offering on-demand dishes and motivating customers to finish their meals by using discounts and rewards. When eating out, people are choosing to order according to their needs and pack up their leftovers. At home, people are making small changes like carefully planning meals and not cooking more than needed.

The “Clean Plate” campaign practises the well-known Chinese tradition of appreciating the value of food. The problem of food waste concerns us all, so ask yourself: am I part of the problem or the solution?

1 What is the “Clean Plate” campaign?

2 What efforts to fight food waste are mentioned in the article?

C In pairs, discuss food waste and think of more ways to fight it. Use the following questions to help you.

- Where does food waste often happen?
- Why should we fight food waste?
- What measures can be taken to fight food waste?

D Write a speech transcript calling on people not to waste food. Use your ideas from part C and the information in parts A and B to help you. Check and improve your writing after you finish it.

Tip

Using a call-to-action statement in persuasive writing

A call-to-action statement is often needed in persuasive writing. It is usually placed at the end of your writing. This statement should be straightforward and state what the writing is all about. It should also be written in a way that will initiate an immediate response from the reader. For example, *Since food waste can have a long-term impact on the environment, let's play a part in helping solve this problem.*

Introduction to food waste

Impacts

Solutions

Conclusion

Extended reading



Read the article from a science magazine about global warming.

Global warming: islands in hot water



The Pacific Ocean covers around one third of the Earth's surface. Scattered across this ocean are thousands of islands. In recent years, the people of these islands have noticed tides getting higher and storms getting worse. And they wonder—is their world about to change? Sadly, the fact is that the islands in the Pacific, especially low-lying coral ones,
5 are suffering the serious effects of global warming.

As global temperatures increase, so do sea levels. Increasing temperatures cause ice sheets and glaciers at the North and South Poles to melt, releasing additional water into the ocean. Warm water also takes up more room than cool water, so as the ocean warms, sea levels rise further. Scientists expect sea levels worldwide may rise by 0.6 to 1.3
10 metres or more by 2100 if the planet continues to warm. On the Pacific Islands, the most obvious sign of rising sea levels is that at high tide, water moves further inland to flow into houses and over roads. In fact, beaches or even whole islands are disappearing: five of the Solomon Islands have already vanished beneath the waves since 1947! Increasing sea levels also contaminate the islands' freshwater supplies. When salty seawater seeps
15 through the ground and gets into wells, the water becomes useless for drinking and irrigation. Besides, growing sea levels interfere with some islands' natural reshaping processes. Usually, waves gradually remove sand from one side of an island and dump new sand on the other side. Unfortunately, however, rising waves accelerate the process, and this certainly causes problems for anyone living on the side that is being washed
20 away.

Meanwhile, increasing global temperatures can have disastrous consequences in other ways. Extreme temperatures generate extreme rainfall patterns. Consequently, the islands go without life-sustaining rainfall for months, leading to crop failure and reduced freshwater reserves. Then suddenly too much comes at once, and precious soil and
25 homes are washed away by the floods. For example, on the hilly Cook Islands, storms have given rise to landslides that carry islanders' homes down the hillsides. Additionally,

the annual tropical cyclones are becoming stronger and happening more frequently, resulting in more severe flooding and landslides. What is more, as the ocean warms, many coral reefs, which act as natural barriers and are home to a rich diversity of ocean species, are dying. Heat and extra carbon dioxide in the atmosphere hurt the corals that build the reefs. Without healthy reefs, fish move away, and the islanders have less to eat.

Despite the challenges, the islanders are not giving up without a fight. For many island communities, fighting climate change is not a choice; it is a matter of survival. They are working with scientists and learning to protect their homes from flooding and storm surges by building sea walls from old bits of coral reefs. They are cutting down fewer trees, since trees hold onto soil. They are also planting mangrove trees to protect the coastal areas. To keep the damaging salt away from their crops, they are growing crops in containers. They are looking for crops that do not mind salty water. Hopefully, they can find a way to stay on their islands, at least for a while.

However, some islanders have made a life-changing decision—moving away from the ocean. It is by no means easy as the ocean has been part of their lives for generations. An entire village on the island of Vanua Levu in Fiji abandoned their coastal homes to move up into the hills. Fortunately, they could begin a new life: buying land, building new houses and growing pineapples. Some miss their old life by the sea, but at least the hills provide refuge from the rising sea, and they are still together. Many of those whose islands have no hills are looking for homes on new islands, or even in new countries. Most do not want to leave, but with the ocean moving in, they feel they have no choice.

No matter what solutions the islanders develop, their islands are still in hot water. Global warming requires not only islanders to fight back. After all, we are living in a human community with a shared future. Maybe we can take a lesson from the ocean. Melting ice in the Arctic makes higher waves in Fiji because all the oceans are connected—there are no walls between them. Change one, change all. If we need to tackle the problem of climate change, we will have to work together as one.

A Write a summary of the science magazine article. Use the following questions to help you.

- What are the effects of global warming on the Pacific Islands?
- What efforts are the islanders making to save their homes and lives?

B What effects of global warming have you witnessed or read about? What could you do to help solve this problem?

Project

Making PPT slides on a problem in sustainable development

A As a class, discuss different problems on our way to achieving sustainable development. You can use the ideas below or think of other ideas. Then in groups, choose one problem to research.

harmful food production practices
ineffective urban planning
overpackaging

poor access to renewable energy
non-sustainable business practices
noise pollution

B As a group, research your chosen problem. Use the ideas below to help you.

- What the problem is
- What effects it has
- What solutions can prevent it

C As a group, put together your information to make your PPT slides. Use the example below to help you. Then present your PPT slides to the rest of the class.

Overfishing

What it is:

Overfishing occurs when so many fish are caught that the number of fish becomes too low.

What effects it has:

- Environmental damage
When the number of some ocean species is being reduced, it creates an imbalance in the ecosystem, further influencing the food web.
- Decreasing food
Many countries rely heavily on fish as their main source of food and overfishing greatly endangers that.

What solutions can prevent it:

- Managing fishing practices better
Governments and communities should work together to create suitable and effective fishing regulations and policies.
- Creating more protected areas
The creation and management of well-planned protected areas can help protect important fish species.
- Improving the supply chain's traceability
Fish importers and sellers label sold fish with information about where the fish came from.



Assessment

Complete the following learning journal to assess your performance. Then work in groups and exchange your learning journal with your partners.

Learning journal

In this unit I learnt _____

I was most interested in _____

I did well in _____

However, there is something I need to improve on: _____

In order to make more progress, I will _____

* Assess your learning of vocabulary and grammar by doing language practice on pages 62–63.

Further study



Our environment has a direct impact on our well-being, so it is essential to protect the environment. Environmental protection involves many possible ideas. Watch a documentary about this topic to learn more and find out how you can do your part.



The world's energy system is shifting its focus towards renewable energy sources, which are clean and sustainable. This change is necessary but also challenging. Read a book about new energy options to learn about the possibilities and problems each one presents.



UNIT 3

Helping people

Love cannot remain by itself—it has no meaning. Love has to be put into action, and that action is service.

—Mother Teresa



In this unit, you are going to:

- read a speech transcript about the United Nations;
- write an article on volunteering;
- read a short story about the value of kindness;
- plan a charity activity.

Welcome to the unit



Helping people in need makes our world a better place to live. Listen to the song below and discuss the following questions in pairs.

Let Our Love Flow Free

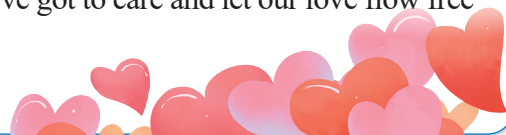
We've got to give if we truly believe
That there are those who need to receive
Some help from another
So no more people suffer
We've got to give and let our love flow free

Now people all over the world
It doesn't matter if you're a boy or a girl
Lend someone a hand
Help them understand
We've got to share and let our love flow free

Now talk is cheap and trouble we've seen
We close our eyes, we're living in between
Knowing what's right, but doing what's wrong
It's time for us all to sing a different song

Take a look in the mirror and see
The person you really need to be
There can be no denying
Too many people are crying
We've got to care and let our love flow free

Now this world's the home that we share
And we really ought to care
To make a better place
For all the human race
We've got to give and let our love flow free
We've got to share and let our love flow free
We've got to care and let our love flow free



- 1 What message does the song try to get across?
- 2 How does this song inspire you to make the world a better place?

Reading



The United Nations helps tackle such problems as war, climate change, poverty and disease on a global scale. The speech below was given by a UN Goodwill Ambassador. Before you read the transcript, think about the following questions:

- What do you know about the United Nations?
- Which world problem do you think requires urgent attention and should be addressed first? Why?

The United Nations— bringing people together



In 1945, the world was on its knees. Cities lay in ruins, economies were destroyed, tens of millions of people were killed, and families across the globe were homeless and mourning the loss of loved ones. World War II, the deadliest human conflict of all time, had shattered people's lives everywhere. From the ashes of this utter destruction rose the
5 organization that we now know as the United Nations (UN), offering hope to a world desperate for lasting peace and security. It is my privilege and pleasure to stand before you today and share with you a little bit about this wonderful organization.

The UN was officially established on 24 October 1945, when the UN Charter, the founding document of the organization, came into force. The following year, the first
10 meeting of the UN was held in London and was attended by representatives of 51 countries. Currently, the UN is made up of over 190 Member States. All of them work together to maintain international peace and security, protect human rights, deliver humanitarian aid, promote sustainable development and uphold international law. Over the years, the UN has been committed to fulfilling its ambition of being the one place
15 where all the world's nations can gather together, discuss common problems, and find shared solutions that benefit all of humanity. In the words of the UN Secretary-General António Guterres, "When international cooperation works, the world wins." The UN believes that international cooperation plays an essential role in addressing many issues, such as clean water, poverty and education.

20 Can you believe that nearly one third of the population around the world did not have access to safe drinking water about a decade ago? The UN has recognized the critical importance of fresh water to life on our planet and recommended that every human being should have between 50 and 100 litres of water per day, for both personal and domestic

uses. With this end in view, the UN has called for transboundary water cooperation
25 through a legal framework for water resources management. Moreover, various water
projects managed by the UN and its partners aim to provide clean water for needy
communities. There has been remarkable progress over the past years. According to
a 2021 UN report, the percentage of the global population with easy access to safely
30 managed drinking water services increased to 74.3%. Increases have been seen in both
rural and urban areas, motivating the UN to continue its tireless work of providing clean
drinking water to those who need it.

With regard to poverty, the UN estimates that around 10% of the world's population,
including one in six children, live in extreme poverty, unable to fulfil their most basic
needs. These figures are both depressing and distressing. Not surprisingly, therefore,
35 ending poverty is the UN's number one Sustainable Development Goal. To effectively
fight poverty, the UN focuses on understanding how people experience poverty in other
ways besides income. Health, education and standard of living are three key dimensions
that can reflect whether people are being left behind. These dimensions comprise ten
indicators, such as nutrition, years of schooling and housing. With this data, the UN
40 can identify the root causes of poverty and work with governments and communities to
ensure that those causes are efficiently addressed.

It is the belief of the UN that education is the most basic building block of every
society and the single best investment countries can make to build prosperous, healthy
and fair societies. However, due to various circumstances like inadequate learning
45 materials or poor school facilities, learning is often difficult for many children. The
United Nations Children's Fund (UNICEF), which is a part of the UN system, has many
ongoing education programmes. One of these, the Reimagine Education initiative, uses
technology to create online learning opportunities, which are accessible to children
anywhere and anytime. Additionally, UNICEF spares no effort to ensure more children
50 have access to their education programmes. These programmes help remove long-
standing barriers to learning, supporting every child's right to learn.

To conclude, I do hope you have found this brief insight into the work of the UN both
inspiring and stimulating. While the UN is a huge global organization, every one of
us can do something to make a positive difference. To quote the famous historian and
55 philosopher Howard Zinn, "Small acts, when multiplied by millions of people, can
transform the world." So I urge you all to find out more about how you can make a
difference to one person, one society or the world, no matter how small.